

# Electronic supplementary material

Payoff scale reshapes how language models play  
the prisoner’s dilemma

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## S1 Scope of the corpus

The analysis in the main text uses one corpus of language-model transcripts, `Dataset/data_fairgame_frontier_llm`, and one synthetic corpus, `Dataset/noise_dataset`, which trains the LSTM branch of the strategy read-out and contains no language-model output at all. No other dataset in the repository enters any number reported in the paper.

The transcript corpus is a complete crossing of ten payoff scales, five models, five prompt languages, four persona pairings and ten replicates, at ten rounds per game and two agents per dyad. Table S1 gives its composition. Every one of the 250 model-by-language-by-scale cells holds exactly 40 dyads, or 200 per cell of the coarser model-by-scale grid, and the ingest raises an error rather than writing its tables if that is not what is on disk.

The completeness guard summarized in the main text has five clauses, and the build refuses to write its tables if any of them fails: a payoff scale is missing; a cell departs from 40 dyads; the cell count is not the product of the factor levels; an agent-game is not ten rounds long; or a model directory was not registered in advance, which is what stops an unrecognised model being

skipped in silence. It also recovers each game’s payoff scale from the payoffs recorded in it, as the median ratio of realised to base payoff, and refuses if that disagrees with the scale the cell was run at by more than a relative  $10^{-6}$ . That last check catches a mislabelled directory.

## S2 Protocol and collection detail

This section carries the collection detail that Materials and methods summarizes: how the models were served, how the prompt files were checked, what the agent was and was not told, and the seeds, parsing and resampling the analysis rests on.

All five models were served through the Kaggle Benchmarks model proxy on an OpenAI-compatible chat-completions endpoint, software development kit version 0.6.1.

The prompt templates are compared byte for byte against the FAIRGAME resource files before any collection run because one save in the wrong encoding corrupts the prompt that reaches the model rather than a comment beside it, and the corruption is invisible in the recorded results.

The opponent’s persona is never disclosed: the protocol gates that disclosure behind a probability, which is set to zero here, and the block naming the opponent is deleted from every prompt.

### S2.1 Seeds

The seed sent with each request is a deterministic function of the base seed 12345 and an index over prompt language, persona pairing and replicate, the round number, and which of the two agents is replying. The index excludes  $\lambda$ , so the ten scales share one stream of seeds. A seed on a hosted endpoint is honoured on a best-effort basis rather than guaranteed, and sampling runs at temperature 1.0, so the scheme is variance reduction and not determinism.

### S2.2 Parsing and fallback

Replies are parsed by the protocol’s own matcher, which normalizes to lower-case alphanumerics and searches in turn for the two option names, for the internal strategy keys, and finally for a bare letter or digit. A reply it cannot resolve is regenerated with a different seed, up to two further attempts, so one decision costs at most three generations. The achieved parse-failure and fallback rates are held in the collection logs rather than in the released per-game tables; each run asserts that its fallback rate is at most 0.02 and fails otherwise.

### S2.3 Resampling and permutation

One seed, 20260905, drives every resampling reported in the paper, so intervals in different rows of a table are drawn against common numbers.

The permutation test shuffles the scale label within strata of prompt language by persona pairing. The unit shuffled is the dyad rather than the agent-game, because the payoff scale was assigned to a whole dyad: under the null both of its agent-games carry the same label, exactly as they do in the data. Shuffling agent-games independently would break that pairing and draw from a null narrower than the true one, which would make the  $p$ -value anti-conservative. It is the same dependence that makes every interval in the paper resample whole dyads and every regression cluster on them. Because the two sides of a mixed pairing are one dyad seen twice,

the strata are the five prompt languages crossed with the three dyad types, cooperative against cooperative, selfish against selfish and the mixed pair, giving 15 strata inside each model: five of 100 dyads, five of 100 and five of 200 respectively.

### S3 Cooperation, by scale, model and language

Table S2 gives the cooperation rate behind figure 1b of the main text, with bootstrap intervals. Table S3 breaks the same quantity down by prompt language and is the table behind figure 2.

The final column of table S3 is the quantity we draw attention to in the main text: the range of cooperation across the ten payoff scales, computed within one model and one language. It runs from 0.080 (Gemini 3.1 Flash-Lite in Arabic) to 0.449 (GPT-5.4 Nano in English). The sensitivity of a model to a transformation that changes nothing about the game is therefore not a single number even for a fixed model; it depends on the language the game was posed in.

### S4 Variance decomposition

Table S4 is the full analysis-of-variance table summarized in the main text. It is computed on the 250 cell means rather than on the 20,000 agent-games, because the design is exactly balanced at the cell level, which makes the type-II decomposition order-independent.

The reading that matters for benchmark practice is the contrast between rows. The payoff scale carries 0.7% of the variance across cells as a main effect and is not significant ( $p = 0.36$ ). Its interaction with the model carries 6.5% and its interaction with the prompt language 3.8%, and both are significant. A design that estimates a main effect for the payoff scale, pooling over models and languages, is therefore built to return exactly the null result it would return if no model responded to the scale at all.

### S5 Persona

Table S5 gives the persona effect at every scale and model, which is the table behind figure 3a. The two leftmost columns are the scales at which every payoff printed is at most 1, since with base payoffs 0, 2, 6, 10 the largest number shown is  $10\lambda$ .

Two features are worth stating explicitly because they are easy to misread from the figure. First, the sign reversal is a reversal of the *persona* effect, not of the cooperation rate: below the boundary the three reversing models cooperate more when told they are selfish than when told they are cooperative. Second, Qwen3 235B-A22B is inverted at every scale rather than only below the boundary, cooperating 0.072 under the cooperative persona against 0.798 under the selfish one. We report it because excluding it would flatter the analysis, but the paper draws no conclusion from the inversion itself beyond noting that its magnitude, like everything else measured here, depends on the payoff scale.

### S6 Strategy read-out

#### S6.1 What the two branches do

Each agent-game is encoded as a sequence of two-letter tokens, one per round, pairing the previous round’s joint outcome as the focal player saw it with the action taken in the current

round; with padding the vocabulary is eleven tokens. The alphabet is shared between the synthetic training corpus and the language-model transcripts, which is what makes a classifier trained on the former applicable to the latter without any adaptation.

The rule base asks an exact question: is this trajectory precisely what AllC, AllD, Tit-for-Tat or Win-Stay-Lose-Shift would have played, given the history that actually occurred? Because all four rules are memory-one, the answer is a per-token lookup and requires no learning. It returns a set rather than a single label, because over ten rounds several canonical rules routinely prescribe the same moves. A player cooperating in all ten rounds against a cooperator is exactly consistent with three of the four rules at once, AllC, Tit-for-Tat and Win-Stay-Lose-Shift, and no method, learned or otherwise, can separate them on that evidence. Reporting the set is the correct answer, not a failure.

The LSTM ranks within whatever set the rule base returns, and decides alone only where the set is empty. It therefore cannot overturn a deduction; it can only break a tie the evidence genuinely leaves open, or supply an attribution where no canonical rule applies.

## S6.2 Provenance, and which claims depend on the classifier

Over the whole corpus, 23.0% of agent-games are matched exactly by one canonical rule, 13.2% by several at once, and 63.7% by none. Those three provenances are reported separately throughout, and the claims in the main text are ordered by how much they depend on the learned component:

- The share of agent-games matched exactly by one rule (table S6) and the number of rounds violating the nearest canonical rule are computed by exact matching. Neither depends on the LSTM in any way. These carry the main text’s strategy claim.
- The compositional claim, that the mix moves from unconditional towards reciprocal play (table S7), uses the labels, and therefore leans on the LSTM for the 76.9% of agent-games that no single rule matches exactly: it decides alone for the 63.7% that match no rule at all, and ranks within the matcher’s own set for the remaining 13.2%. It is the weaker of the two and is presented as such.

Of the agent-games carrying the AllC label, 26.0% were deduced from an exact rule match and a further 29.6% came from a rule set the LSTM ranked within; for AllD the deduced share is 32.7%.

Table S8 shows that this dependence is very unevenly spread across the four labels, which bounds how the compositional result may be read. AllC and AllD carry deduced shares of 26.0% and 32.7%, so a movement in those two is anchored in part by exact matching. Tit-for-Tat and Win-Stay-Lose-Shift are 95.4% and 97.3% unmatched, meaning the label is an attribution the LSTM makes about which canonical rule the play most resembles, on trajectories that follow none of them exactly. A rise in the Tit-for-Tat share is therefore evidence that play is drifting towards the reciprocal region of the space, and is not evidence that any model is implementing Tit-for-Tat. The main text states this where the result is reported.

## S6.3 Validation

Table S9 reports the accuracy of the LSTM branch. Training uses the noise-free and 5% execution-noise variants of the synthetic corpus together, balanced at 40,320 sequences per strategy per level on a random 80/20 split, which forces the network to identify a rule from

imperfect behaviour rather than from a clean signature; the 10% and 20% variants are held out entirely and measure how the read-out degrades as play departs further from any rule.

The network is a single-layer LSTM with 24-dimensional embeddings, 96 hidden units and dropout 0.15, read at its last valid step. The optimiser is AdamW at learning rate  $3 \times 10^{-3}$  and weight decay  $10^{-4}$ , with a cosine schedule over 12 epochs, batches of 512, gradient-norm clipping at 1.0 and seed 1234.

Accuracy falls from 98.5% in distribution to 93.6% and 81.1% at those two unseen noise levels, which is the honest bound on how far the label distributions in table S7 should be pushed.

## S7 Why the model curves cancel under pooling

Table S10 gives the pairwise correlations between the five models’ cooperation-against-scale curves. Six of the ten pairs are negative, the mean is  $-0.10$ , and the extremes are  $-0.72$  and  $+0.73$ . Averaging curves that disagree in shape this strongly reduces the range from a mean of 0.136 within a model to 0.048 for the average, a factor of 2.8. The cancellation is arithmetic rather than statistical, and no increase in sample size affects it.

## S8 Reproduction

The pipeline runs in order and each stage writes its outputs under `Analysis/scaling/`:

- `s00_build.py` parses the transcripts into `rounds.parquet` and `games.parquet`, and refuses to write either if the grid is incomplete.
- `s01_train_lstm.py` trains the LSTM branch on the synthetic corpus and writes `models/strategy_lstm.pt`.
- `s02_readout.py` applies the hybrid read-out to all 20,000 agent-games and writes `readout.parquet`.
- `s03_stats.py` and `s04_strategy_stats.py` compute every statistic in the paper into `tables/`.
- `s05_figures.py` draws figures 1 to 6.
- `s06_tables.py` and `s07_supplementary.py` generate the LaTeX tables in the main text and in this supplement.
- `s08_verify_paper.py` re-derives every number quoted in the prose of the main text from the tidy tables and fails if any disagrees.

Random seeds are fixed: 1234 for classifier training, 20260905 for the bootstrap and the permutation test.

## S9 Supplementary tables

**Table S1: The corpus, by model.** Every model was run on the same ten payoff scales and five prompt languages. “Cell  $n$ ” is the number of dyads in each model  $\times$  language  $\times$  scale cell; the design is balanced with none missing, and the build refuses to write its tables otherwise.

| Model                 | Scales | Lang. | Dyads  | Agent-games | Decisions | Cell $n$ | Coop. |
|-----------------------|--------|-------|--------|-------------|-----------|----------|-------|
| Claude Haiku 4.5      | 10     | 5     | 2,000  | 4,000       | 40,000    | 40       | 0.415 |
| GPT-5.4 Nano          | 10     | 5     | 2,000  | 4,000       | 40,000    | 40       | 0.606 |
| Gemini 3.1 Flash-Lite | 10     | 5     | 2,000  | 4,000       | 40,000    | 40       | 0.617 |
| Gemini 3.5 Flash-Lite | 10     | 5     | 2,000  | 4,000       | 40,000    | 40       | 0.751 |
| Qwen3 235B-A22B       | 10     | 5     | 2,000  | 4,000       | 40,000    | 40       | 0.435 |
| <b>All</b>            | 10     | 5     | 10,000 | 20,000      | 200,000   | 40       | 0.565 |

**Table S2: Cooperation by model and payoff scale.** Each entry aggregates 200 dyads. The bracketed line under each model is a 95% confidence interval from a bootstrap that resamples whole dyads.

| Model                 | $\lambda=0.01$       | $\lambda=0.1$        | $\lambda=0.25$       | $\lambda=0.5$        | $\lambda=1$          | $\lambda=2$          | $\lambda=5$          | $\lambda=10$         | $\lambda=100$        | $\lambda=1000$       |
|-----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Claude Haiku 4.5      | 0.407<br>[0.37,0.45] | 0.379<br>[0.34,0.42] | 0.371<br>[0.34,0.40] | 0.427<br>[0.39,0.46] | 0.452<br>[0.42,0.49] | 0.444<br>[0.41,0.48] | 0.424<br>[0.39,0.46] | 0.435<br>[0.40,0.47] | 0.407<br>[0.38,0.43] | 0.401<br>[0.37,0.43] |
| GPT-5.4 Nano          | 0.438<br>[0.40,0.48] | 0.540<br>[0.50,0.58] | 0.644<br>[0.61,0.68] | 0.598<br>[0.56,0.63] | 0.576<br>[0.54,0.61] | 0.637<br>[0.60,0.67] | 0.680<br>[0.65,0.71] | 0.633<br>[0.60,0.67] | 0.663<br>[0.64,0.69] | 0.653<br>[0.62,0.68] |
| Gemini 3.1 Flash-Lite | 0.641<br>[0.61,0.67] | 0.648<br>[0.62,0.68] | 0.621<br>[0.59,0.65] | 0.640<br>[0.61,0.67] | 0.635<br>[0.61,0.67] | 0.614<br>[0.58,0.65] | 0.614<br>[0.58,0.65] | 0.598<br>[0.56,0.63] | 0.584<br>[0.54,0.62] | 0.577<br>[0.54,0.61] |
| Gemini 3.5 Flash-Lite | 0.792<br>[0.74,0.84] | 0.694<br>[0.64,0.74] | 0.872<br>[0.85,0.89] | 0.800<br>[0.77,0.83] | 0.774<br>[0.75,0.80] | 0.710<br>[0.67,0.74] | 0.697<br>[0.66,0.73] | 0.754<br>[0.72,0.78] | 0.734<br>[0.70,0.77] | 0.682<br>[0.64,0.72] |
| Qwen3 235B-A22B       | 0.476<br>[0.43,0.52] | 0.494<br>[0.44,0.54] | 0.455<br>[0.41,0.50] | 0.432<br>[0.39,0.47] | 0.416<br>[0.38,0.45] | 0.427<br>[0.39,0.46] | 0.419<br>[0.38,0.45] | 0.422<br>[0.39,0.45] | 0.399<br>[0.37,0.43] | 0.411<br>[0.38,0.44] |

**Table S3: Cooperation by model, prompt language and payoff scale.** 40 dyads per entry. The final column is the range across the ten scales for that model and language; it runs from 0.080 to 0.449, so the scale sensitivity of a model is a property of the model and the language jointly.

| Model                 | Language   | 0.01 | 0.1  | 0.25 | 0.5  | 1    | 2    | 5    | 10   | 100  | 1000 | Range        |
|-----------------------|------------|------|------|------|------|------|------|------|------|------|------|--------------|
| Claude Haiku 4.5      | English    | 0.35 | 0.40 | 0.32 | 0.42 | 0.44 | 0.45 | 0.41 | 0.42 | 0.45 | 0.46 | <b>0.136</b> |
|                       | French     | 0.65 | 0.47 | 0.50 | 0.33 | 0.37 | 0.43 | 0.41 | 0.40 | 0.38 | 0.40 | <b>0.314</b> |
|                       | Vietnamese | 0.17 | 0.17 | 0.28 | 0.29 | 0.26 | 0.20 | 0.30 | 0.22 | 0.23 | 0.23 | <b>0.130</b> |
|                       | Chinese    | 0.53 | 0.52 | 0.45 | 0.61 | 0.69 | 0.57 | 0.56 | 0.66 | 0.58 | 0.56 | <b>0.245</b> |
|                       | Arabic     | 0.35 | 0.33 | 0.31 | 0.48 | 0.51 | 0.56 | 0.43 | 0.47 | 0.40 | 0.36 | <b>0.249</b> |
| GPT-5.4 Nano          | English    | 0.45 | 0.64 | 0.85 | 0.88 | 0.86 | 0.90 | 0.88 | 0.87 | 0.89 | 0.76 | <b>0.449</b> |
|                       | French     | 0.75 | 0.78 | 0.78 | 0.57 | 0.55 | 0.60 | 0.62 | 0.67 | 0.55 | 0.74 | <b>0.226</b> |
|                       | Vietnamese | 0.49 | 0.46 | 0.58 | 0.47 | 0.50 | 0.62 | 0.75 | 0.54 | 0.65 | 0.63 | <b>0.289</b> |
|                       | Chinese    | 0.24 | 0.45 | 0.65 | 0.60 | 0.49 | 0.57 | 0.61 | 0.58 | 0.62 | 0.65 | <b>0.414</b> |
|                       | Arabic     | 0.25 | 0.37 | 0.37 | 0.48 | 0.47 | 0.49 | 0.54 | 0.51 | 0.61 | 0.49 | <b>0.360</b> |
| Gemini 3.1 Flash-Lite | English    | 0.74 | 0.62 | 0.67 | 0.62 | 0.63 | 0.62 | 0.59 | 0.62 | 0.57 | 0.62 | <b>0.167</b> |
|                       | French     | 0.57 | 0.66 | 0.70 | 0.63 | 0.65 | 0.60 | 0.60 | 0.61 | 0.60 | 0.56 | <b>0.145</b> |
|                       | Vietnamese | 0.58 | 0.56 | 0.54 | 0.64 | 0.57 | 0.55 | 0.57 | 0.49 | 0.47 | 0.47 | <b>0.170</b> |
|                       | Chinese    | 0.62 | 0.66 | 0.53 | 0.61 | 0.59 | 0.60 | 0.60 | 0.58 | 0.58 | 0.58 | <b>0.133</b> |
|                       | Arabic     | 0.69 | 0.74 | 0.66 | 0.70 | 0.74 | 0.69 | 0.71 | 0.69 | 0.69 | 0.66 | <b>0.080</b> |
| Gemini 3.5 Flash-Lite | English    | 0.94 | 0.66 | 0.92 | 0.84 | 0.74 | 0.59 | 0.53 | 0.65 | 0.64 | 0.60 | <b>0.412</b> |
|                       | French     | 0.91 | 0.63 | 0.86 | 0.72 | 0.78 | 0.63 | 0.75 | 0.78 | 0.59 | 0.61 | <b>0.317</b> |
|                       | Vietnamese | 0.72 | 0.63 | 0.93 | 0.86 | 0.81 | 0.76 | 0.62 | 0.64 | 0.89 | 0.73 | <b>0.314</b> |
|                       | Chinese    | 0.69 | 0.80 | 0.76 | 0.64 | 0.70 | 0.73 | 0.67 | 0.76 | 0.66 | 0.73 | <b>0.162</b> |
|                       | Arabic     | 0.70 | 0.75 | 0.89 | 0.94 | 0.83 | 0.84 | 0.92 | 0.93 | 0.89 | 0.74 | <b>0.242</b> |
| Qwen3 235B-A22B       | English    | 0.47 | 0.51 | 0.53 | 0.44 | 0.42 | 0.49 | 0.43 | 0.50 | 0.37 | 0.46 | <b>0.155</b> |
|                       | French     | 0.41 | 0.40 | 0.35 | 0.39 | 0.37 | 0.36 | 0.35 | 0.35 | 0.34 | 0.33 | <b>0.083</b> |
|                       | Vietnamese | 0.52 | 0.55 | 0.48 | 0.48 | 0.45 | 0.45 | 0.47 | 0.44 | 0.40 | 0.42 | <b>0.150</b> |
|                       | Chinese    | 0.48 | 0.52 | 0.46 | 0.44 | 0.43 | 0.45 | 0.45 | 0.44 | 0.49 | 0.47 | <b>0.084</b> |
|                       | Arabic     | 0.50 | 0.50 | 0.46 | 0.42 | 0.40 | 0.40 | 0.40 | 0.38 | 0.39 | 0.38 | <b>0.124</b> |

**Table S4: Variance decomposition of the cell means.** Type-II analysis of variance on the 250 model  $\times$  language  $\times$  scale cell means, where the design is exactly balanced. The payoff scale has no significant main effect while both of its interactions are significant.

| Term                    | Sum sq. | df  | <i>F</i> | <i>p</i> | % variance |
|-------------------------|---------|-----|----------|----------|------------|
| Model                   | 3.9188  | 4   | 206.80   | < 0.001  | 56.5       |
| Language                | 0.2092  | 4   | 11.04    | < 0.001  | 3.0        |
| Payoff scale            | 0.0471  | 9   | 1.10     | 0.364    | 0.7        |
| Model $\times$ language | 1.3665  | 16  | 18.03    | < 0.001  | 19.7       |
| Model $\times$ scale    | 0.4542  | 36  | 2.66     | < 0.001  | 6.5        |
| Language $\times$ scale | 0.2615  | 36  | 1.53     | 0.041    | 3.8        |
| Residual                | 0.6822  | 144 |          |          | 9.8        |

**Table S5: Persona effect by model and payoff scale.** The cooperation rate of an agent instructed that it is cooperative minus that of an agent instructed that it is selfish, at each scale. The two leftmost columns are the scales at which every payoff printed is at most 1.

| Model                 | $\lambda=0.01$ | $\lambda=0.1$ | $\lambda=0.25$ | $\lambda=0.5$ | $\lambda=1$ | $\lambda=2$ | $\lambda=5$ | $\lambda=10$ | $\lambda=100$ | $\lambda=1000$ |
|-----------------------|----------------|---------------|----------------|---------------|-------------|-------------|-------------|--------------|---------------|----------------|
| Claude Haiku 4.5      | -0.211         | -0.232        | +0.206         | +0.200        | +0.179      | +0.261      | +0.277      | +0.158       | +0.168        | +0.211         |
| GPT-5.4 Nano          | -0.172         | -0.098        | +0.015         | -0.026        | -0.053      | +0.032      | +0.041      | +0.055       | -0.017        | +0.100         |
| Gemini 3.1 Flash-Lite | +0.552         | +0.602        | +0.640         | +0.670        | +0.693      | +0.681      | +0.707      | +0.696       | +0.659        | +0.692         |
| Gemini 3.5 Flash-Lite | -0.335         | -0.578        | +0.122         | +0.234        | +0.241      | +0.258      | +0.313      | +0.203       | +0.193        | +0.230         |
| Qwen3 235B-A22B       | -0.901         | -0.922        | -0.811         | -0.768        | -0.702      | -0.635      | -0.690      | -0.602       | -0.657        | -0.566         |

**Table S6: Share of agent-games matched exactly by one canonical rule (%).** Computed by exact rule matching, with no learned component. 400 agent-games per entry.

| Model                 | $\lambda=0.01$ | $\lambda=0.1$ | $\lambda=0.25$ | $\lambda=0.5$ | $\lambda=1$ | $\lambda=2$ | $\lambda=5$ | $\lambda=10$ | $\lambda=100$ | $\lambda=1000$ |
|-----------------------|----------------|---------------|----------------|---------------|-------------|-------------|-------------|--------------|---------------|----------------|
| Claude Haiku 4.5      | 20.0           | 15.8          | 3.2            | 3.5           | 4.0         | 2.8         | 4.0         | 2.5          | 4.5           | 3.5            |
| GPT-5.4 Nano          | 12.5           | 7.0           | 9.2            | 6.2           | 5.5         | 6.5         | 8.2         | 7.2          | 8.5           | 9.0            |
| Gemini 3.1 Flash-Lite | 16.5           | 19.0          | 21.2           | 23.5          | 29.0        | 23.0        | 30.0        | 29.5         | 27.5          | 26.8           |
| Gemini 3.5 Flash-Lite | 30.8           | 33.8          | 20.2           | 13.0          | 20.5        | 14.2        | 14.8        | 11.5         | 14.0          | 16.2           |
| Qwen3 235B-A22B       | 77.0           | 67.2          | 63.0           | 62.7          | 59.8        | 45.2        | 58.0        | 43.2         | 55.0          | 42.0           |



**Table S7: Strategy composition by model and payoff scale (%).** The label assigned by the hybrid read-out: the rule base fixes the candidate set and the LSTM ranks within it, deciding alone only where no canonical rule fits. Columns sum to 100 within each model and scale.

| Model                 | Label | 0.01 | 0.1  | 0.25 | 0.5  | 1    | 2    | 5    | 10   | 100  | 1000 |
|-----------------------|-------|------|------|------|------|------|------|------|------|------|------|
| Claude Haiku 4.5      | AllC  | 27.0 | 20.0 | 14.2 | 20.0 | 21.8 | 18.2 | 15.2 | 18.8 | 12.8 | 14.0 |
|                       | TFT   | 20.5 | 23.5 | 23.5 | 18.0 | 16.8 | 16.5 | 19.5 | 17.8 | 20.8 | 22.2 |
|                       | WSLS  | 8.8  | 13.5 | 15.2 | 13.5 | 20.2 | 21.2 | 20.5 | 18.2 | 21.8 | 19.0 |
|                       | AllD  | 43.8 | 43.0 | 47.0 | 48.5 | 41.2 | 44.0 | 44.8 | 45.2 | 44.8 | 44.8 |
| GPT-5.4 Nano          | AllC  | 29.8 | 41.8 | 52.2 | 46.2 | 48.5 | 54.2 | 58.5 | 52.2 | 53.2 | 50.7 |
|                       | TFT   | 11.8 | 11.5 | 10.5 | 11.8 | 10.2 | 13.2 | 13.8 | 13.2 | 13.0 | 17.8 |
|                       | WSLS  | 12.5 | 12.5 | 14.5 | 15.5 | 13.8 | 10.5 | 12.5 | 10.2 | 15.5 | 14.0 |
|                       | AllD  | 46.0 | 34.2 | 22.8 | 26.5 | 27.5 | 22.0 | 15.2 | 24.2 | 18.2 | 17.5 |
| Gemini 3.1 Flash-Lite | AllC  | 55.8 | 57.0 | 58.2 | 61.0 | 59.8 | 55.0 | 58.5 | 53.8 | 52.2 | 52.2 |
|                       | TFT   | 6.5  | 3.8  | 2.2  | 0.2  | 2.5  | 3.5  | 1.2  | 2.2  | 6.2  | 5.8  |
|                       | WSLS  | 12.8 | 13.2 | 8.8  | 2.8  | 8.0  | 7.2  | 6.5  | 7.2  | 7.2  | 6.0  |
|                       | AllD  | 25.0 | 26.0 | 30.8 | 36.0 | 29.8 | 34.2 | 33.8 | 36.8 | 34.2 | 36.0 |
| Gemini 3.5 Flash-Lite | AllC  | 73.8 | 59.8 | 74.0 | 64.0 | 58.0 | 60.0 | 51.7 | 59.8 | 56.2 | 51.5 |
|                       | TFT   | 6.8  | 2.2  | 10.2 | 16.2 | 17.0 | 16.2 | 15.2 | 13.5 | 15.2 | 14.2 |
|                       | WSLS  | 6.2  | 13.2 | 14.0 | 16.2 | 21.2 | 12.0 | 18.0 | 18.5 | 16.5 | 15.2 |
|                       | AllD  | 13.2 | 24.8 | 1.8  | 3.5  | 3.8  | 11.8 | 15.0 | 8.2  | 12.0 | 19.0 |
| Qwen3 235B-A22B       | AllC  | 42.0 | 47.8 | 39.2 | 38.0 | 36.8 | 36.5 | 35.5 | 35.2 | 34.5 | 33.8 |
|                       | TFT   | 0.5  | 0.0  | 0.0  | 0.0  | 1.2  | 0.8  | 2.0  | 6.2  | 0.0  | 4.8  |
|                       | WSLS  | 9.0  | 2.5  | 8.8  | 13.5 | 7.2  | 12.0 | 8.5  | 8.2  | 9.2  | 10.0 |
|                       | AllD  | 48.5 | 49.8 | 52.0 | 48.5 | 54.8 | 50.7 | 54.0 | 50.2 | 56.2 | 51.5 |

**Table S8: Provenance of each label.** For every label, the share of the agent-games carrying it that were deduced (exactly one canonical rule fits), ambiguous (several fit, and the LSTM ranked within that set) or unmatched (no canonical rule fits, so the label is an attribution the LSTM makes alone). The asymmetry is severe and bounds how the compositional result may be read: AllC and AllD carry a substantial deduced share, whereas the TFT and WSLS labels are almost entirely attributions.

| Label | $n$   | Deduced (%) | Ambiguous (%) | Unmatched (%) |
|-------|-------|-------------|---------------|---------------|
| AllC  | 8,884 | 26.0        | 29.6          | 44.4          |
| TFT   | 2,010 | 4.6         | 0.0           | 95.4          |
| WSLS  | 2,494 | 2.0         | 0.7           | 97.3          |
| AllD  | 6,612 | 32.7        | 0.0           | 67.3          |

**Table S9: Validation of the LSTM branch.** Trained on synthetic AllC / AllD / Tit-for-Tat / Win-Stay-Lose-Shift trajectories at zero and 5% execution noise, balanced at 40,320 sequences per strategy per level. The 10% and 20% noise levels are never trained on. No language-model transcript enters training.

| Split                         | $n$     | Accuracy |
|-------------------------------|---------|----------|
| held-out (NoNoise + Noise005) | 64,512  | 0.9846   |
| OOD (Noise01)                 | 161,280 | 0.9364   |
| OOD (Noise02)                 | 161,280 | 0.8107   |
| held-out recall: AllC         | 16,150  | 0.9853   |
| held-out recall: AllD         | 16,203  | 0.9952   |
| held-out recall: TFT          | 16,193  | 0.9744   |
| held-out recall: WSLS         | 15,966  | 0.9834   |

**Table S10: Pairwise correlation between the model curves.** Correlation of the ten-point cooperation-against-scale curves of each pair of models. Six of the ten pairs are negative and the mean is  $-0.10$ , which is why averaging the curves cancels most of the movement.

|                       | Claude Haiku | GPT-5.4 Nano | Gemini 3.1 | Gemini 3.5 | Qwen3 235B-A22B |
|-----------------------|--------------|--------------|------------|------------|-----------------|
| Claude Haiku 4.5      |              | +0.11        | -0.03      | -0.18      | -0.56           |
| GPT-5.4 Nano          | +0.11        |              | -0.70      | -0.23      | -0.72           |
| Gemini 3.1 Flash-Lite | -0.03        | -0.70        |            | +0.36      | +0.73           |
| Gemini 3.5 Flash-Lite | -0.18        | -0.23        | +0.36      |            | +0.22           |
| Qwen3 235B-A22B       | -0.56        | -0.72        | +0.73      | +0.22      |                 |